

Animate Asset Replacer

Recipe JSON Reference - v0.3.2

Reference for the key/value pairs used by the recipe JSON for embedded image and audio dataURI replacement. Relative replacement paths are resolved from the directory containing the recipe JSON.

1. Recipe Structure

A recipe contains global defaults plus a replacements array. Settings on an individual replacement override the corresponding global settings.

```
{
  "task": "replace_embedded_manifest_assets",
  "image_processing": { ... },
  "operation": { ... },
  "conditions": { ... },
  "verification": { ... },
  "replacements": [
    { ... },
    { ... }
  ],
  "output": {
    "type": "html",
    "filename_suffix": "_assets_replaced"
  }
}
```

2. Top-Level Keys

Key	Value / Type	Applies To	Meaning
task	string	Recipe	Use "replace_embedded_manifest_assets" for mixed image/audio recipes. Older image-only values remain supported.
image_processing	object	Images	Global image-processing defaults. Individual image entries can override these.
operation	object	Images / Audio	Global encoding/output defaults. Individual replacements can override these.
conditions	object	Images / Audio	Global conditions evaluated for each replacement.
verification	object	Images	Global verification options.
replacements	array	Images / Audio	One object per manifest asset to replace.
output	object	Recipe	Controls the generated HTML filename.

3. Replacement Entry Keys

Key	Value / Type	Applies To	Meaning
-----	--------------	------------	---------

asset_id	string; required	Images / Audio	Manifest id to locate, e.g. "bg" or "spinSfx".
asset_type	"image" "audio"	Images / Audio	Explicitly identifies the replacement type. Recommended, especially for audio.
replacement_image	string path	Images	Replacement image file. Relative paths resolve from the recipe JSON directory.
replacement_audio	string path	Audio	Replacement audio file. Relative paths resolve from the recipe JSON directory.
image_processing	object	Images	Per-image overrides for global image_processing.
operation	object	Images / Audio	Per-asset overrides for global operation.
conditions	object	Images / Audio	Per-asset conditions, combined with global conditions.
verification	object	Images	Per-image verification overrides.

4. Image Processing Keys

Key	Value / Type	Applies To	Meaning
match_original_image_dimensions	boolean	Images	When true, resize the replacement to the dimensions of the original embedded image.
resize_mode	"stretch" "contain" "cover"	Images	Controls how Sharp fits the replacement when resizing. stretch maps to fill.
preserve_aspect_ratio	boolean	Images	Documented recipe option. Current resize behavior is primarily controlled by resize_mode.
skip_resize_if_dimensions_already_match	boolean	Images	Avoids an unnecessary resize when source and original dimensions already match.
allow_cropping	boolean	Images	Documented recipe option; current processing behavior is governed by resize_mode.
allow_padding	boolean	Images	Documented recipe option; current processing behavior is governed by resize_mode.
target_width	integer or string	Images	Accepted by the schema; current v0.3.2 processor does not directly use this key.
target_height	integer or string	Images	Accepted by the schema; current v0.3.2 processor does not directly use this key.

5. Operation Keys

Key	Value / Type	Applies To	Meaning
preserve_replacement_format	boolean	Images	When true, keep the replacement image format when supported (PNG/JPEG/WebP).
mime_type	MIME string	Images / Audio	Explicit output MIME. Images

			commonly use "image/jpeg" or "image/png"; audio can use "audio/mpeg", "audio/wav", etc.
jpeg_quality	integer 1-100	Images	JPEG output quality. Defaults to 95 when not specified.
extract_original_image_dimensions	boolean	Images	Accepted recipe option; original dimensions are inspected automatically by the current processor.
resize_replacement_before_encoding	boolean	Images	Accepted recipe option; current resize behavior is controlled by image_processing.
data_uri_source	string	Images / Audio	Accepted recipe option; current processor obtains dataURI from the matched manifest entry.
convert_replacement_image_to_data_uri	boolean	Images	Accepted recipe option; conversion to Base64 dataURI is automatic.
replace_target_image_data	boolean	Images	Accepted recipe option; replacement is automatic once conditions pass.

6. Audio Behavior

Audio files are not transcoded in v0.3.2. The replacement file bytes are read as-is and embedded into the HTML as Base64. MIME type is selected from operation.mime_type when supplied; otherwise the app infers it from the file extension, falling back to the original embedded MIME type.

Key	Value / Type	Applies To	Meaning
.mp3 / .mpeg / .mpga	audio/mpeg	Audio	Automatically inferred MIME.
.wav	audio/wav	Audio	Automatically inferred MIME.
.ogg / .oga	audio/ogg	Audio	Automatically inferred MIME.
.m4a / .mp4	audio/mp4	Audio	Automatically inferred MIME.
.aac	audio/aac	Audio	Automatically inferred MIME.
.webm	audio/webm	Audio	Automatically inferred MIME.

7. Conditions

Key	Value / Type	Applies To	Meaning
asset_id_equals	string	Images / Audio	Only replace when the current manifest asset id equals this value.
asset_type_equals	"image" "audio"	Images / Audio	Only replace when the detected/declared asset type matches.
replacement_width_greater_than	number	Images	Replacement image width must be greater than this value.
replacement_width_at_least	number	Images	Replacement image width must be at least this value.
replacement_height_at_least	number	Images	Replacement image height must be at least this value.
replacement_format_in	array of strings	Images	Allowed Sharp image formats, e.g. ["png", "jpeg"].
original_width_equals	number	Images	Original embedded image width must equal this value.
original_height_equals	number	Images	Original embedded image height must equal this value.
replacement_audio_bytes_at	number	Audio	Replacement audio file size

_least			must be at least this many bytes.
replacement_audio_extension_in	array of strings	Audio	Allowed extensions, including the leading dot, e.g. [".mp3", ".wav"].

8. Verification Keys

Key	Value / Type	Applies To	Meaning
verify_encoded_image_dimensions_match_original	boolean	Images	After encoding, verifies that output image width/height match the original embedded image.

9. Output Keys

Key	Value / Type	Applies To	Meaning
type	"html"	Recipe	Output document type.
filename_suffix	string	Recipe	Appended to the source HTML filename, e.g. "_assets_replaced".

10. Image Example

```
{
  "task": "replace_embedded_manifest_assets",
  "image_processing": {
    "match_original_image_dimensions": true,
    "resize_mode": "stretch",
    "skip_resize_if_dimensions_already_match": true
  },
  "operation": {
    "preserve_replacement_format": true
  },
  "replacements": [
    {
      "asset_id": "bg",
      "asset_type": "image",
      "replacement_image": "new_bg.jpg"
    },
    {
      "asset_id": "logo",
      "asset_type": "image",
      "replacement_image": "images/new_logo.png",
      "operation": {
        "preserve_replacement_format": true
      },
      "conditions": {
        "replacement_width_at_least": 200
      }
    }
  ],
  "output": {
    "type": "html",
    "filename_suffix": "_images_replaced"
  }
}
```

11. Audio Example

```
{
  "task": "replace_embedded_manifest_assets",
  "replacements": [
    {
      "asset_id": "spinSfx",
      "asset_type": "audio",
      "replacement_audio": "audio/spin.mp3",
      "operation": {
        "mime_type": "audio/mpeg"
      }
    },
    {
      "asset_id": "winSfx",
      "asset_type": "audio",
      "replacement_audio": "audio/win.wav",
      "conditions": {
        "replacement_audio_extension_in": [".wav"]
      }
    }
  ],
  "output": {
    "type": "html",
    "filename_suffix": "_audio_replaced"
  }
}
```

12. Mixed Image + Audio Example

```
{
  "task": "replace_embedded_manifest_assets",
  "image_processing": {
    "match_original_image_dimensions": true,
    "resize_mode": "stretch"
  },
  "replacements": [
    {
      "asset_id": "bg",
      "asset_type": "image",
      "replacement_image": "new_bg.jpg"
    },
    {
      "asset_id": "spinSfx",
      "asset_type": "audio",
      "replacement_audio": "spin.mp3",
      "operation": {
        "mime_type": "audio/mpeg"
      }
    }
  ],
  "output": {
    "type": "html",
    "filename_suffix": "_assets_replaced"
  }
}
```

13. Custom JavaScript Conditions

The file `src/rules/conditions.js` runs once per replacement after JSON conditions. Return true to perform the replacement, false to skip it, or throw an Error to stop processing.

```
async function shouldReplace(context) {
  if (
    context.asset.type === "audio" &&
    context.replacementAudio.bytes > 5 * 1024 * 1024
  ) {
    throw new Error("Replacement audio must be 5 MB or smaller.");
  }

  return true;
}
```

14. Path Resolution

A relative path such as `"new_bg.jpg"`, `"audio/spin.mp3"`, or `"images/logo.png"` is resolved relative to the folder containing the selected recipe JSON. Absolute paths are used unchanged.

15. Notes on Current v0.3.2 Implementation

The schema accepts a few legacy/configuration keys that are not independently acted upon by the current processor. These are identified in the tables above. The keys that directly control current behavior are `asset_id`, `asset_type`, `replacement_image/replacement_audio`, `image_processing.match_original_image_dimensions`, `resize_mode`, `skip_resize_if_dimensions_already_match`, `operation.preserve_replacement_format`, `operation.mime_type`, `operation.jpeg_quality`, the implemented conditions, verification, and `output.filename_suffix`.